ML Engineer

Probability Prediction System

Seiichi Tanaka
STEM Engineer | AI Orchestrator
30 Years Education | 5AI Integrated

Price Reversal Zone | Mixed Density Network | 155 Features | 8-Week POC

1,850+
Live Trades Executed

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8 Weeks
POC Delivery Timeline

WHY ME: Trader Who Codes

**"A trader who codes,
not a coder who trades."**

Your 155 features = my daily trading toolkit.

Competitive Edge

Typical ML Engineer	Seiichi Tanaka
Financial data = just data	**1,850+ trades = lived experience**
Must learn feature meaning	**Uses these features daily**
Reversal = math concept	**Reversal = actual entry point**
Domain ramp-up needed	**Day-1 domain value delivery**
PyTorch expert, no trading	**PyTorch learning + 100p spec**

Trading Track Record

1,850+
Total Trades

Win Rate

Technical Stack

Core Python | Pandas | NumPy | FastAPI

ML / DL PyTorch | TensorFlow | MDN

Trading Pine Script | MQL5 | TradingView API

Data PostgreSQL | Time-series | Feature Eng.

AI Ops Claude + GPT-4 + Gemini + Grok + Manus

8-Week POC Roadmap

1 **Week 1-2** — **Spec Deep-Dive + Feature Pipeline**

Complete 100+ page specification comprehension.
Build 155-feature extraction and validation pipeline.

2 **Week 3-4** — **MDN Architecture Implementation**

Dual regime prediction model (P80 + P20).
Initial training and evaluation on historical data.

3 **Week 5-6** — **Walk-Forward Validation**

Training pipeline optimization and tuning.
Hyperparameter search with cross-validation.

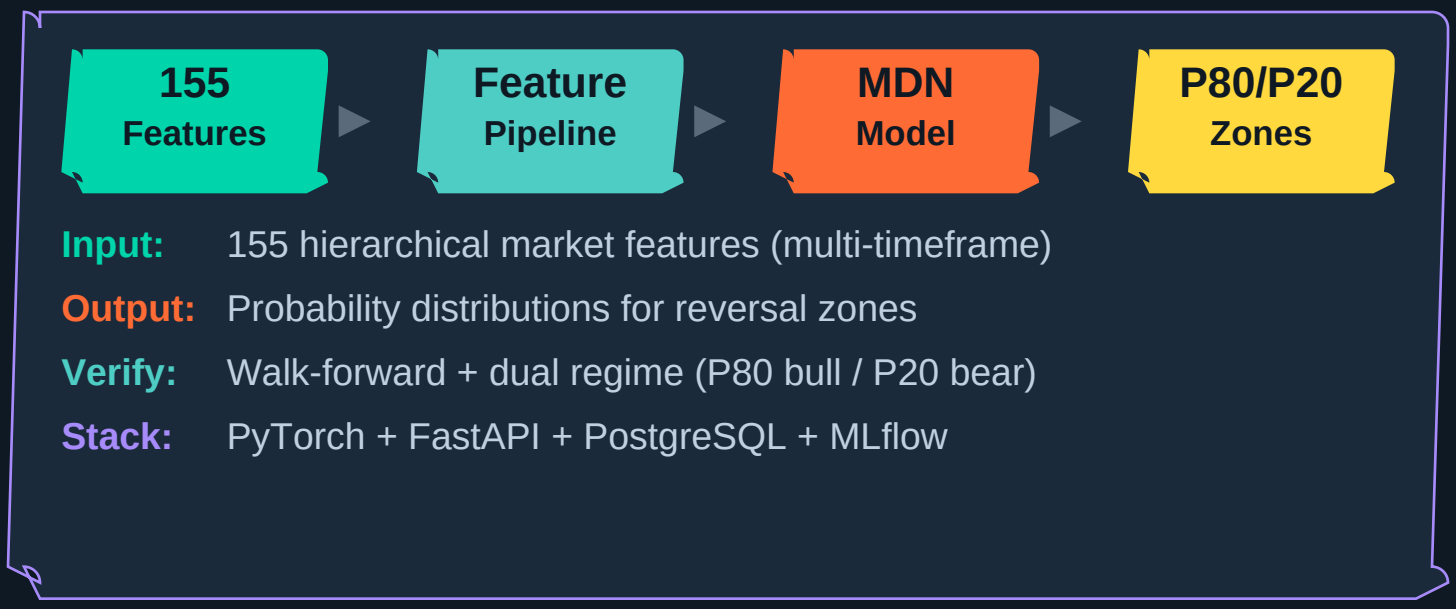
4 **Week 7** — **Inference API + Model Versioning**

FastAPI endpoint with intelligent caching layer.
MLflow model version and experiment management.

5 **Week 8** — **Documentation + Handoff**

Comprehensive technical documentation.
Deployment guide and clean code handoff.

MDN Architecture



Input: 155 hierarchical market features (multi-timeframe)

Output: Probability distributions for reversal zones

Verify: Walk-forward + dual regime (P80 bull / P20 bear)

Stack: PyTorch + FastAPI + PostgreSQL + MLflow

5AI Orchestration

Claude	GPT-4	Gemini	Grok	Manus
Code Review	Prototyping	Data Analysis	Research	Automation

Deliverables

- ✓ Trained MDN model with walk-forward validation results
- ✓ FastAPI inference endpoint with intelligent caching
- ✓ Feature extraction pipeline (155 features)
- ✓ MLflow model versioning & experiment tracking
- ✓ Full documentation & deployment guide